

(MR22-1CS0110) DATA ANALYTICS

III Year B. Tech – II Semester

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COURSE OBJECTIVES:

- ❖ Learning about the Importance of Data and its importance
- ❖ Knowing Python fundamentals and Pandas essentials
- ❖ Learning the Principles of Probability and sampling Methods
- ❖ Getting knowledge about formulating and testing hypothesis
- ❖ Learning and analytical comparison's with ANOVA methods
- ❖ Learning about Performance indicators using ROC methods

UNIT-III

Hypothesis testing: Importance of Hypothesis testing, null and alternative hypotheses, Type-I and Type –II errors, approaches to Hypothesis testing, two sample testing

Reference Books:

1. McKinney, W. (2012). Python for data analysis: Data wrangling with Pandas, NumPy, and IPython. " O'Reilly Media, Inc.".
2. Swaroop, C. H. (2003). A Byte of Python. Python Tutorial.
3. Ken Black, sixth Editing. Business Statistics for Contemporary Decision Making. “John Wiley & Sons, Inc”.
4. Anderson Sweeney Williams (2011). Statistics for Business and Economics. “Cengage Learning”.

❖ Hypothesis Testing

- ❖ Hypothesis Testing is a statistical method used to make decisions or inferences about a population based on sample data. It involves formulating an assumption (hypothesis), testing it using appropriate statistical techniques, and determining whether the assumption is likely to be true or false.
- ❖ *Hypothesis testing is a systematic procedure for deciding whether the results of a research study support a particular theory which applies to a population*
- ❖ **Example: A school introduces a new teaching method and claims that it improves students' average exam scores. To test this, the scores of students taught using the new method are compared to those taught using the traditional method.**
- ❖ It provides a systematic framework for evaluating research hypotheses and determining the likelihood of observed results occurring due to chance.
- ❖ A hypothesis enables researchers not only to discover a relationship between variables, but also to predict a relationship based on theoretical guidelines and/or empirical evidence.

❖ **Advantages of Hypothesis Testing:**

- ❖ Hypothesis testing provides an objective, systematic way to make decisions based on data rather than intuition or subjective judgment.
- ❖ It helps remove biases from the decision-making process.

General Information

❖ **Null Hypothesis (H_0) and Alternative Hypothesis (H_a / H_1).**

- ❖ The **null hypothesis** represents the **default or conservative (traditional) position**, assuming **no significant difference or effect in the population**.
- ❖ The **alternative hypothesis** **contradicts** the null hypothesis and suggests that there is a **significant difference or effect**.

❖ **Examples:**

- ❖ **Null Hypothesis: H_0 : There is no difference in the salary of factory workers based on gender.**
- ❖ **Alternative Hypothesis: H_a : Male factory workers have a higher salary than female factory workers.**
- ❖ NOTE: You want to reject the null hypothesis, but how and when can you do that?
- ❖ To start, you'll need to perform a statistical test on your data. Then has to decide the Null Hypothesis is rejected.

Process of Hypothesis Testing

The process of hypothesis testing involves the following steps:

- ❖ Hypothesis testing is a formal procedure for investigating our ideas about the world using statistics. It is most often used by scientists to test specific predictions, called hypotheses, that arise from theories.

There are 5 main steps in hypothesis testing:

1. State your research hypothesis as a null hypothesis (H_0) and alternate hypothesis (H_a or H_1).
2. Collect data in a way designed to test the hypothesis.
3. Perform an appropriate statistical test.
4. Decide whether to reject or fail to reject your null hypothesis.
5. Present the findings in your results and discussion section.

Step 1: State your null and alternate hypothesis

- ❖ After developing your initial research hypothesis (the prediction that you want to investigate), it is important to restate it as a null (H_0) and alternate (H_a) hypothesis so that you can test it mathematically.
- ❖ The alternate hypothesis is usually your initial hypothesis that predicts a relationship between variables. The null hypothesis is a prediction of no relationship between the variables you are interested in.

Step 2: Collect data

- ❖ For a statistical test to be valid, it is important to perform sampling and collect data in a way that is designed to test your hypothesis.
- ❖ If your data are not representative, then you cannot make statistical inferences about the population you are interested in.

Step 3: Perform a statistical test

- ❖ There are a variety of statistical tests available, but they are all based on the comparison of **within-group variance** (how spread out the data is within a category) **versus** **between-group variance** (how different the categories are from one another).
- ❖ If the **between-group variance** is substantial enough to result in minimal or no overlap between groups, your statistical test will indicate this with a **low p-value**.
- ❖ This implies that the observed differences between these groups are unlikely to have occurred by chance.
- ❖ Alternatively, If there is **high within-group variance** and **low between-group variance**, your statistical test will be characterized by a **high p-value**.
- ❖ This suggests that any observed difference between groups is likely attributable to chance.
- ❖ The selection of your statistical test should be guided by the nature of variables and the level of measurement in your collected data.

Step 4: Decide whether to reject or fail to reject your null hypothesis

- ❖ Based on the outcome of your statistical test, you will have to decide whether to reject or fail to reject your null hypothesis.
- ❖ In most cases you will use the p-value generated by your statistical test to guide your decision. And in most cases, your predetermined level of significance for rejecting the null hypothesis will be 0.05 – that is, when there is a less than 5% chance that you would see these results if the null hypothesis were true.
- ❖ In some cases, researchers choose a more conservative level of significance, such as 0.01 (1%). This minimizes the risk of incorrectly rejecting the null hypothesis (Type I error).

Step 5: Present your findings

- ❖ The results of hypothesis testing will be presented in the results and discussion sections of your research paper, dissertation or thesis.
- ❖ In the results section you should give a brief summary of the data and a summary of the results of your statistical test (for example, the estimated difference between group means and associated p-value). In the discussion, you can discuss whether your initial hypothesis was supported by your results or not.
- ❖ In the formal language of hypothesis testing, we talk about rejecting or failing to reject the null hypothesis. You will probably be asked to do this in your statistics assignments.

Example:

Given that the average height of men is 175.4 cm, the average height of women is 161.7 cm, and the p-value from a one-tailed t-test is 0.002 with a significance level of 0.05, what conclusion can be drawn about the null hypothesis that men are not taller than women?

1. Formulating the Hypotheses:

- ❖ **Null Hypothesis (H_0):** Men are not taller than women, on average. (i.e., the average height of men is less than or equal to that of women).
- ❖ **Alternative Hypothesis (H_a):** Men are taller than women, on average. (i.e., the average height of men is greater than that of women).
- ❖ Since you expect men to be taller, this is a one-tailed test.

2. Data Collection:

- ❖ **Sample Requirements:** Equal representation of men and women, diversity in socio-economic classes, and controlling for variables that might influence height (e.g., age, diet, region).
- ❖ **Scope:** Decide whether the study is localized (e.g., for one country) or global.
- ❖ Example: Census data is ideal as it covers a wide range of demographic groups.

Step 3: Perform a statistical test

- ❖ You perform a one-tailed t-test to compare the mean heights of men and women.
- ❖ The t-test provides :
- ❖ Estimate of the difference in means: 13.7 cm. (175.4 cm - 161.7 cm)
- ❖ P-value: 0.002, which indicates the probability of observing such a difference if the null hypothesis (no difference) were true.
- ❖ The confidence interval for the difference is from 10.2 cm to infinity, meaning the difference is statistically significant and not due to random chance.

Step 4: Decide whether to reject or fail to reject your null hypothesis

- ❖ Significance Level (α): The standard cutoff is 0.05.
- ❖ Since the p-value (0.002) is less than 0.05, you reject the null hypothesis (H_0).
- ❖ Conclusion: The data strongly supports the alternative hypothesis (H_a), suggesting that men are, on average, taller than women.

Step 5: Present your findings

- ❖ In comparing the mean heights of men and women, we found an average difference of 13.7 cm, with a p-value of 0.002. This indicates that the difference is statistically significant.
- ❖ Therefore, we reject the null hypothesis and conclude that men are, on average, taller than women

Understanding T-values and P-values

- ❖ The P-value and T-value are important concepts in statistical hypothesis testing. They are used to assess whether the results of an analysis are statistically significant.
- ❖ **T-Value (Test Statistic):**
- ❖ The T-value is a calculated value that measures the **difference between the observed data and what is expected under the null hypothesis**, expressed in terms of standard errors.
- ❖ It is part of the t-test, which is used to compare the means of groups or to assess the significance of a coefficient in a regression model.
- ❖ **Formula for T-value, For a one-sample t-test:**

$$t = \frac{\bar{x} - \mu}{s / \sqrt{n}}$$

Where:

- ❖ $\bar{x}$: Sample mean
- ❖ μ : Population mean under the null hypothesis
- ❖ s : Standard deviation of the sample
- ❖ n : Sample size

Understanding:

- ❖ A large absolute T-value indicates a large difference between the observed data and the null hypothesis.
- ❖ The sign of the T-value indicates the direction of the difference.

P-Value

- ❖ The P-value is the probability of observing a test statistic (like the T-value) as extreme as, or more extreme than, the observed value, assuming the null hypothesis is true.
- ❖ **Key Points about P-Value:**
 - ❖ **Range:** The P-value is between 0 and 1.
 - ❖ **Threshold for significance:** A common threshold is 0.05 (5%).
 - ❖ If $P \leq 0.05$: Reject the null hypothesis (results are statistically significant).
 - ❖ If $P > 0.05$: Fail to reject the null hypothesis (results are not statistically significant).
- ❖ **Example:** If the P-value is 0.03, it means there is a 3% probability that the observed data could occur by random chance under the null hypothesis.

Relationship Between T-Value and P-Value

- ❖ T-value helps determine how far the sample statistic (e.g., sample mean) is from the null hypothesis.
- ❖ The P-value is derived from the T-value using the t-distribution. It quantifies the evidence against the null hypothesis.

T-Value (Magnitude)	P-Value Range	Significance	Interpretation
Large (High Positive or Negative)	$P < 0.01$	Very Significant	Strong evidence to reject the null hypothesis; the observed difference is highly unlikely due to chance.
Moderate	$0.01 \leq P < 0.05$	Significant	Sufficient evidence to reject the null hypothesis; observed difference is unlikely due to chance.
Small	$0.05 \leq P < 0.1$	Marginally Significant	Weak evidence; borderline case where the result may be considered significant depending on the context.
Close to Zero	$P \geq 0.1$	Not Significant	No sufficient evidence to reject the null hypothesis; the observed difference is likely due to chance.
Direction (Sign)	N/A	Depends on Hypothesis	Positive T-value indicates one direction of effect; negative T-value indicates the opposite direction.

Given that the average height of men is 175.4 cm, the average height of women is 161.7 cm, and the p-value from a one-tailed t-test is 0.002 with a significance level of 0.05, what conclusion can be drawn about the null hypothesis that men are not taller than women?

Importing the Required Library

1. `import scipy.stats as stats`

2. Data Setup

1. `men_heights = [175.4] # Example: Average height of men in cm`
2. `women_heights = [161.7] # Example: Average height of women in cm`
3. `n_men = 100 # Example sample size for men`
4. `n_women = 100 # Example sample size for women`
5. `std_men = 10 # Assumed standard deviation of men heights`
6. `std_women = 9 # Assumed standard deviation of women heights`

3. Compute the t-statistic and P-value for a One-Tailed Test

1. $\text{mean_diff} = \text{men_heights}[0] - \text{women_heights}[0]$

❖ This calculates the difference in the average heights between men and women.

1. $\text{pooled_std} = ((\text{std_men}^2 / n_{\text{men}}) + (\text{std_women}^2 / n_{\text{women}}))^{0.5}$

❖ This calculates the pooled standard deviation. The formula used here combines the standard deviations of both groups, taking into account the sample sizes. The pooled standard deviation is a measure of the spread of data from both groups combined.

1. $t_{\text{stat}} = \text{mean_diff} / \text{pooled_std}$

❖ This computes the t-statistic, which measures how many standard deviations the observed difference in means (height difference) is from the expected difference under the null hypothesis (assuming no difference in means). The formula for the t-statistic is the difference in means divided by the pooled standard error.

1. `df = n_men + n_women - 2 # Degrees of freedom`

❖ This calculates the degrees of freedom (df), which is used to determine the distribution of the t-statistic. For independent samples, it's the total sample size minus 2.

1. `p_value = stats.t.sf(t_stat, df) # One-tailed p-value`

❖ This calculates the p-value for a one-tailed test using the cumulative distribution function (CDF) of the t-distribution. sf is the survival function, which gives the probability of obtaining a t-statistic greater than or equal to the calculated t-statistic under the null hypothesis.

❖ **Displaying the Results**

1. `alpha = 0.05 # Significance level`

2. `print(f"T-Statistic: {t_stat:.3f}")`

3. `print(f"P-Value: {p_value:.3f}")`

❖ This sets the significance level alpha to 0.05 (a common threshold for statistical significance). Then, it prints the computed t-statistic and p-value, formatted to three decimal place

Decision Making

1. if $p_value < \alpha$:
2. `print("Reject the null hypothesis: Men are likely taller than women.")`
3. else:
4. `print("Fail to reject the null hypothesis: Insufficient evidence to claim men are taller than women.")`

❖ **Based on the p-value, the program decides whether to reject the null hypothesis**

- ❖ If the p-value is smaller than the significance level α (0.05), it indicates strong evidence against the null hypothesis, so the program will reject it and conclude that men are likely taller than women.
- ❖ If the p-value is greater than α , there's not enough evidence to reject the null hypothesis, so it concludes there's insufficient evidence to claim that men are taller than women.

Importance of Hypothesis Testing

- ❖ According to the San Jose State University Statistics Department, hypothesis testing is one of the most important concepts in statistics because it is how you decide if something really happened,
- ❖ or if certain treatments have positive effects,
- ❖ or if groups differ from each other
- ❖ or if one variable predicts another.
- ❖ **In short, you want to proof if your data is statistically significant and unlikely to have occurred by chance alone. In essence then, a hypothesis test is a test of significance.**
- ❖ Hypothesis testing allows the researcher to determine whether the data from the sample is statistically significant.
- ❖ Hypothesis testing is one of the most important processes for measuring the validity and reliability of outcomes in any systematic investigation

❖ **Here are some key reasons why hypothesis testing is important:**

❖ **Testing Research Questions:**

- ❖ Hypothesis testing helps researchers test specific research questions or hypotheses.
- ❖ By formulating clear hypotheses, researchers can make predictions about the relationships between variables or the effects of interventions.
- ❖ Hypothesis testing provides a framework to evaluate these predictions objectively.

❖ **Statistical Inference:**

- ❖ Hypothesis testing enables researchers to draw conclusions about a population using sample data.
- ❖ By analyzing the sample data, researchers can determine whether the observed results are likely to occur due to chance or if they represent a true effect in the population.
- ❖ This allows for generalization from a smaller sample to a larger population.

❖ **Decision-Making:**

- ❖ Hypothesis testing provides a structured decision-making process. Researchers can define a null hypothesis (H_0), which represents no effect or no difference, and an alternative hypothesis (H_1), which represents an effect or difference.
- ❖ By comparing the sample data to the null hypothesis, researchers can make decisions such as rejecting the null hypothesis in favor of the alternative or failing to reject the null hypothesis due to insufficient evidence.

❖ **Objectivity and Reliability:**

- ❖ Hypothesis testing offers an objective approach to evaluating research hypotheses.
- ❖ It allows researchers to rely on statistical evidence rather than personal opinions or biases.
- ❖ The procedures followed in hypothesis testing are well-defined and can be replicated by other researchers, promoting transparency and the advancement of scientific knowledge.

❖ Null Hypotheses

❖ Definition of the Null Hypothesis

- ❖ The Null Hypothesis (H_0) is a fundamental concept in statistics that represents the default assumption in hypothesis testing.
- ❖ It states that there is no effect, no difference, or no relationship between variables in a population.
- ❖ It is the hypothesis that a researcher seeks to test and possibly reject.
- ❖ In simple terms, the null hypothesis assumes that any observed effect in a sample is due to random chance rather than a real effect in the population.
- ❖ **Structure:** Typically, H_0 is a statement of equality (e.g., $\mu = \mu_0$, where μ is a population mean)

❖ **Purpose of the Null Hypothesis**

- ❖ It provides a benchmark against which researchers test their alternative hypothesis.
- ❖ It helps in decision-making regarding the acceptance or rejection of new theories.
- ❖ It ensures that conclusions are not drawn based on random variations in data.

❖ **Representation of the Null Hypothesis**

- ❖ The null hypothesis is usually denoted as H_0 and formulated mathematically based on the type of test being conducted.
- ❖ Example 1: Testing a Population Mean
 - ❖ $H_0: \mu = \mu_0$ (The population mean is equal to a specific value μ_0)
- ❖ Example 2: Testing a Difference Between Two Groups
 - ❖ $H_0: \mu_1 = \mu_2$ (There is no difference between the means of two groups)
- ❖ Example 3: Testing Correlation Between Two Variables
 - ❖ $H_0: \rho = 0$ (There is no correlation between the two variables)

❖ Importance of the Null Hypothesis

- ❖ It helps in establishing a scientific and objective approach to hypothesis testing.
- ❖ It prevents researchers from making false claims without statistical evidence.
- ❖ It ensures that findings are statistically significant and not due to random chance.

❖ Testing the Null Hypothesis

- ❖ The null hypothesis is tested using statistical hypothesis testing methods. The key steps include:
 - ❖ State the Null (H_0) and Alternative Hypothesis (H_1).
 - ❖ Choose a significance level (α), typically 0.05 or 0.01.
 - ❖ Collect sample data and perform a statistical test, such as:
 - ❖ t-test
 - ❖ Chi-square test
 - ❖ ANOVA
 - ❖ Regression analysis

Alternative Hypothesis (H_1)

❖ Definition of the Alternative Hypothesis (H_1)

- ❖ The Alternative Hypothesis (H_1 or H_a) is the opposite of the Null Hypothesis (H_0) in statistical hypothesis testing.
- ❖ It states that there is a significant effect, difference, or relationship between variables in the population.
- ❖ If the null hypothesis is rejected, the alternative hypothesis is considered to be supported by the data.
- ❖ In simple terms, the alternative hypothesis represents what a researcher wants to prove or expects to be true.

❖ Purpose of the Alternative Hypothesis

- ❖ It suggests the presence of an effect, difference, or relationship in the population.
- ❖ It helps researchers test new theories, treatments, or interventions.
- ❖ It provides direction for further research and decision-making.

❖ **Representation of the Alternative Hypothesis**

- ❖ The alternative hypothesis is usually denoted as H_1 or H_a and is formulated based on the type of hypothesis test being conducted.
- ❖ Example 1: Testing a Population Mean
 - ❖ $H_1: \mu \neq \mu_0$ (The population mean is different from μ_0)
- ❖ Example 2: Testing a Difference Between Two Groups
 - ❖ $H_1: \mu_1 \neq \mu_2$ (There is a difference between the means of two groups)
- ❖ Example 3: Testing Correlation Between Two Variables
 - ❖ $H_1: \rho \neq 0$ (There is a correlation between the two variables)
- ❖ Example 4: Testing Proportions
 - ❖ $H_1: p_1 \neq p_2$ (The proportions of two groups are not equal)

❖ **Types of Alternative Hypothesis**

❖ The type of alternative hypothesis depends on the research question and statistical test:

❖ Two-Tailed Alternative Hypothesis ($H_1: \mu \neq \mu_0$)

❖ Suggests that the population parameter is different from a certain value (it can be higher or lower).

❖ Example: "The average height of students in this school is different from 170 cm."

❖ Used in tests where deviations in both directions matter.

❖ One-Tailed Alternative Hypothesis ($H_1: \mu > \mu_0$ or $H_1: \mu < \mu_0$)

❖ Suggests that the population parameter is either greater than or less than a certain value.

❖ Example (Right-tailed test): "Students who study for more than 5 hours score higher than those who don't."

❖ Example (Left-tailed test): "The new drug reduces blood pressure more than the placebo."

❖ Used in cases where only one direction is relevant.

❖ **Common Errors in Hypothesis Testing – NULL Hypothesis**

- ❖ Type I Error (False Positive): Rejecting H_0 when it is actually true.
- ❖ Type II Error (False Negative): Failing to reject H_0 when it is actually false.

❖ **Common Errors in Hypothesis Testing – Alternative Hypothesis**

- ❖ Type I Error (False Positive): Rejecting H_0 when it is actually true (incorrectly supporting H_1).
- ❖ Type II Error (False Negative): Failing to reject H_0 when it is actually false (failing to support H_1).

TYPE-I ERRORS AND TYPE –II ERRORS

- ❖ When performing hypothesis testing, two possible errors can occur: Type I Error (False Positive) and Type II Error (False Negative). These errors arise because statistical tests make decisions based on sample data, which may not always perfectly represent the entire population.

False Positive

- ❖ A False Positive occurs when a test incorrectly indicates the presence of a condition when it is actually absent.
- ❖ In Hypothesis Testing: A False Positive happens when we reject a true null hypothesis (H_0)—this is also called a Type I Error.
- ❖ **Examples of False Positive**
- ❖ **Medical Testing:** A COVID-19 test wrongly detects the virus in a healthy person.
- ❖ **Spam Filter:** A legitimate email is mistakenly marked as spam.
- ❖ **Security Systems:** A motion sensor falsely triggers an alarm when no intruder is present.
- ❖ **Courtroom Decision:** An innocent person is wrongly convicted of a crime.

❖ **Impact of False Positive**

- ❖ Can lead to unnecessary actions (e.g., unnecessary medical treatment).
- ❖ Wastes resources and creates inconvenience.
- ❖ Can reduce trust in a system (e.g., unreliable security alerts).

❖ **False Negative**

- ❖ A False Negative occurs when a test fails to detect a condition that is actually present.
- ❖ **In Hypothesis Testing:** A False Negative happens when we fail to reject a false null hypothesis (H_0)—this is also called a Type II Error.

❖ **Examples of False Negative**

- ❖ **Medical Testing:** A pregnancy test fails to detect pregnancy in a pregnant woman.
- ❖ **Spam Filter:** A phishing email is mistakenly classified as a normal email.
- ❖ **Security Systems:** A facial recognition system fails to detect a known criminal.
- ❖ **Courtroom Decision:** A guilty person is mistakenly acquitted.

❖ Impact of False Negative

- ❖ Can lead to missed opportunities for action (e.g., undiagnosed diseases).
- ❖ May cause serious consequences, especially in medical or security-related scenarios.
- ❖ Often more dangerous than False Positives in critical applications.

❖ False Positive vs. False Negative

Aspect	False Positive (Type I Error)	False Negative (Type II Error)
Definition	Incorrectly detecting a condition that is not present.	Failing to detect a condition that is actually present.
Hypothesis Testing Equivalent	Type I Error (Rejecting a true H_0)	Type II Error (Failing to reject a false H_0)
Medical Example	A healthy person is diagnosed with a disease.	A sick person is told they are healthy.
Spam Filter Example	A valid email is marked as spam.	A spam email is classified as normal.

❖ How to Reduce False Positives and False Negatives

❖ Balancing Sensitivity and Specificity

- ❖ Sensitivity: Reduces False Negatives (detects actual cases).

- ❖ Specificity: Reduces False Positives (avoids incorrect detections).

❖ Improving Test Accuracy (e.g., better medical tests, stronger AI models).

❖ Setting Thresholds Intelligently

- ❖ In spam filters, a higher threshold reduces False Positives but increases False Negatives.

❖ Using Multiple Tests or Approaches (e.g., confirming a medical test with a second test).

$$\text{Sensitivity} = \frac{\text{True Positives (TP)}}{\text{True Positives (TP)} + \text{False Negatives (FN)}}$$

$$\text{Specificity} = \frac{\text{True Negatives (TN)}}{\text{True Negatives (TN)} + \text{False Positives (FP)}}$$

❖ **Type I error (False Positive):**

- ❖ A Type I error occurs when the null hypothesis (H_0) is incorrectly rejected, even though it is actually true.
- ❖ In other words, it is the error of claiming a relationship or effect exists when, in reality, there is no significant evidence to support that claim.
- ❖ It represents a false positive result. The probability of committing a Type I error is denoted as alpha (α), which is the predetermined significance level in hypothesis testing.
- ❖ A lower significance level reduces the likelihood of Type I errors but increases the chance of Type II errors.

❖ **Note:**

- ❖ **A test result that indicates that a person has a specific disease or condition when the person actually does not have the disease or condition**

- ❖ The type I error significance level or rate level is the probability of refusing the null hypothesis given that it is true.
- ❖ It is represented by Greek letter α (alpha) and is also known as alpha level. Usually, the significance level or the probability of type I error is set to 0.05 (5%), assuming that it is satisfactory to have a 5% probability of inaccurately rejecting the null hypothesis.

❖ **Example:**

- ❖ In a clinical trial, the null hypothesis might be that a new drug has no effect, while the alternative hypothesis suggests that the drug is effective.
- ❖ Committing a Type I error would mean concluding that the drug is effective when, in reality, it has no significant impact.

❖ **Formula for Type I Error Probability**

- ❖ The probability of making a Type I Error is denoted by α (alpha), also called the significance level of a test.

$$\text{Type I Error Rate} = \alpha$$

Common α values used in hypothesis testing:

- ❖ 0.05 (5%) → There is a 5% chance of making a Type I Error.
- ❖ 0.01 (1%) → There is a 1% chance of making a Type I Error
- ❖ A lower α reduces the likelihood of a Type I Error but increases the chance of a Type II Error.
- ❖ Examples of Type I Error (False Positive)
- ❖ **Medical Diagnosis Example**
 - ❖ Scenario: A COVID-19 test incorrectly identifies a healthy person as infected.
 - ❖ Error: The test falsely detects the virus when the person is actually healthy.
 - ❖ Impact: The person may receive unnecessary medication or quarantine.

❖ **Type II error (False Negative):**

- ❖ A Type II error occurs when the null hypothesis (H_0) is erroneously accepted, regardless of the alternative hypothesis (H_1) being true.
- ❖ It is the error of failing to claim a relationship or effect that does exist.
- ❖ It represents a false negative result. The probability of committing a Type II error is denoted as beta (β).
- ❖ The complement of beta, known as the power of the test, is equal to $(1 - \beta)$ and represents the ability to correctly reject the null hypothesis when it is false.

❖ **Note:**

A false positive is an outcome where the model incorrectly predicts the positive class. And a false negative is an outcome where the model incorrectly predicts the negative class

❖ **Example:**

- ❖ Scenario: A medical test is designed to detect a disease, and the null hypothesis is that the patient is healthy.
- ❖ Type II Error: The test fails to detect the disease in a patient who actually has it, leading to a false negative result.

❖ **Example Scenario:**

- ❖ Consider a medical test designed to detect a specific disease, let's call it Disease X. The hypotheses in this scenario are:
- ❖ Null Hypothesis (H_0): The patient does not have Disease X.
- ❖ Alternative Hypothesis (H_1): The patient has Disease X.
- ❖ Type II Error in Detail:
- ❖ If the test result fails to detect Disease X in a patient who actually has it, this is a Type II error.

❖ **Outcome:**

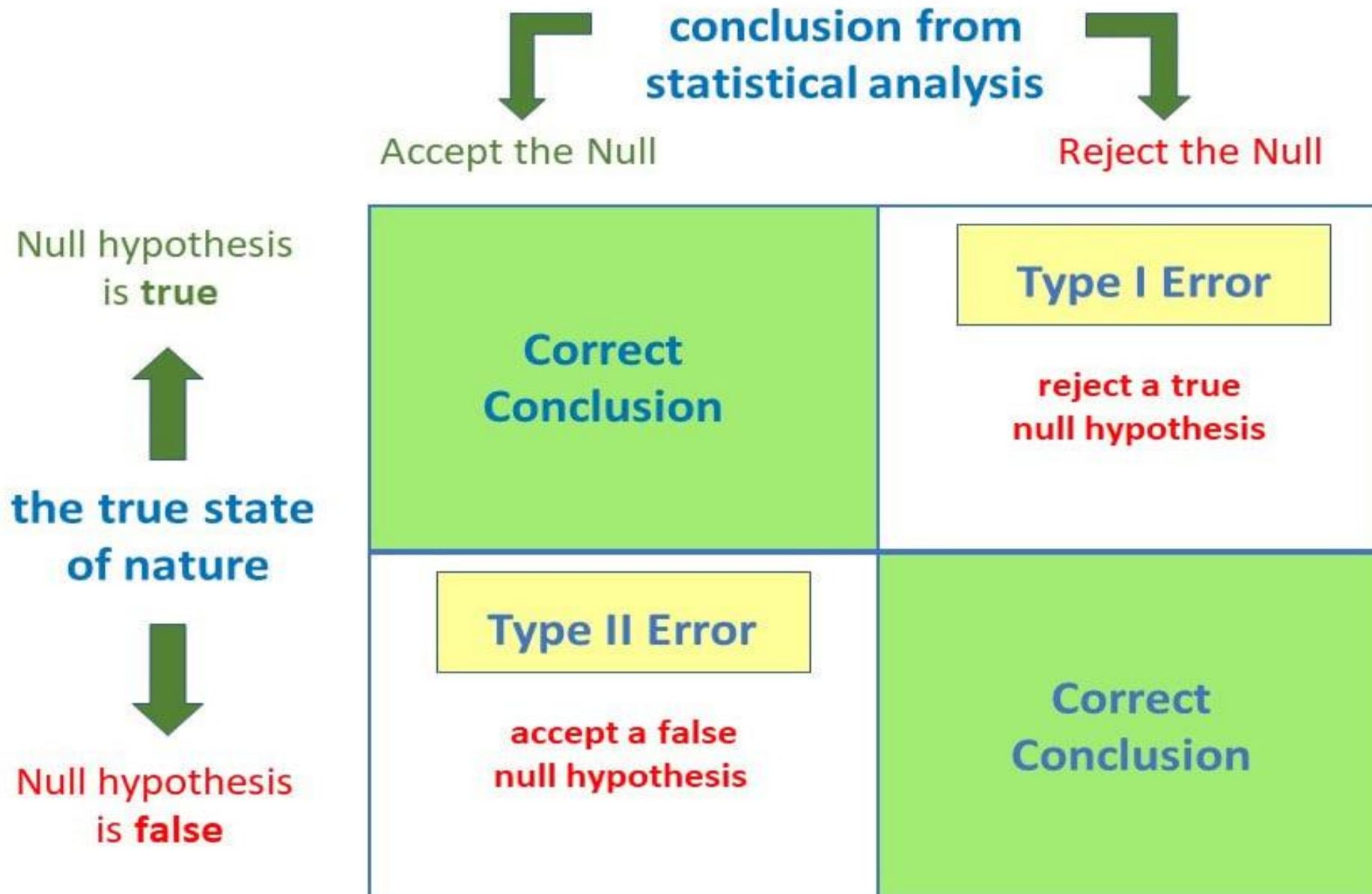
- ❖ Test Result: Negative for Disease X.
- ❖ Reality: The patient actually has Disease X.

❖ **Implications and Consequences:**

- ❖ A Type II error in medical testing can have significant consequences, as it means the test has failed to identify a real condition. Some key points to understand: Missed Diagnoses, Impact on Patient Health,

❖ **Minimizing Type II Errors:**

- ❖ **Increasing Sensitivity:** Improving the sensitivity of the test is crucial to reduce the risk of Type II errors. This may involve refining the testing methodology or using more sensitive diagnostic tools.
- ❖ **Validation and Calibration:** Rigorous validation studies are essential to ensure that the test performs accurately across diverse populations and conditions.
- ❖ **Clinical Judgment:** Healthcare professionals often rely on a combination of test results and clinical judgment to make informed decisions, especially when there is suspicion of a condition despite a negative test result.
- ❖ **Type II errors in medical testing is critical for healthcare professionals and researchers to enhance the accuracy and reliability of diagnostic procedures, ultimately improving patient outcomes.**



Identify and explain the different approaches to hypothesis testing.

Approaches to Hypothesis Testing:

1. Formulating Hypotheses:

- ❖ Hypothesis testing starts with formulating the null hypothesis (H_0) and the alternative hypothesis (H_1).
- ❖ The null hypothesis represents the assumption of no effect or no relationship, while the alternative hypothesis suggests the presence of an effect or relationship.

2. Setting the Significance Level:

- ❖ The significance level (alpha, α) is predetermined before conducting the test and represents the maximum probability of making a Type I error.
- ❖ Commonly used significance levels are 0.05 (5%) and 0.01 (1%).
- ❖ The choice of significance level depends on the field of study, consequences of errors, and desired level of confidence.

3. Selecting a Statistical Test:

- ❖ Based on the research question and characteristics of the data, an appropriate statistical test is chosen.
- ❖ Common tests include t-tests, chi-square tests, ANOVA, regression analysis, and more.

4. Analyzing the Data:

- ❖ Researchers collect a sample of data and perform the chosen statistical test.
- ❖ The test produces a test statistic and a p-value, which are used to make a decision regarding the null hypothesis.

5. Comparing p-value and Significance Level:

- ❖ The p-value represents the probability of obtaining the observed data or more extreme data assuming the null hypothesis is true.
- ❖ If the p-value is less than or equal to the significance level ($p \leq \alpha$), we reject the null hypothesis.
- ❖ If the p-value is greater than the significance level ($p > \alpha$), we fail to reject the null hypothesis.

6. Interpreting the Results:

- ❖ If the null hypothesis is rejected, we may conclude that there is a significant effect or relationship.
- ❖ If the null hypothesis is not rejected, we do not have sufficient evidence to support the alternative hypothesis.
- ❖ **It is essential to strike a balance between Type I and Type II errors by appropriately selecting the significance level, considering sample size calculations, using suitable statistical methods, and conducting independent replications of studies.**
- ❖ **The choice of significance level depends on the specific context, consequences of errors, and desired level of confidence.**

CHI-SQUARE TEST

- ❖ A chi-square test is a statistical test that is used to **compare observed and expected results**.
- ❖ A chi-square test is a statistical hypothesis test that surveys whether two categorical variables are independent in influencing the test statistic.
- ❖ It's used to compare observed results with expected results to determine if a difference is due to chance or a relationship between the variables.
- ❖ **The goal of this test is to identify whether a inequality between actual and predicted data is due to chance or to a link between the variables under consideration.**
- ❖ A chi-square test or comparable nonparametric test is required to test a hypothesis regarding the distribution of a categorical variable.
- ❖ Categorical variables, which indicate categories such as animals or countries, can be nominal or ordinal. They cannot have a normal distribution since they can only have a few particular values.

- ❖ It is used to calculate the difference between two categorical variables, which are:
 - ❖ As a result of chance
 - ❖ Because of the relationship
- ❖ Formula For Chi-Square Test

$$\chi^2_c = \frac{\sum (O_i - E_i)^2}{E_i}$$

Where

c = Degrees of freedom

O = Observed Value

E = Expected Value

- ❖ The degrees of freedom in a statistical calculation represent the number of variables that can vary in a calculation. The degrees of freedom can be calculated to ensure that chi-square tests are statistically valid. These tests are frequently used to compare observed data with data that would be expected to be obtained if a particular hypothesis were true.
- ❖ The Observed values are those you gather yourselves.
- ❖ The expected values are the frequencies expected, based on the null hypothesis.

Use of Chi-Square Test:

- ❖ Here are some of the uses of the Chi-Squared test:
- ❖ The Chi-squared test can be used to see if your data follows a well-known theoretical probability distribution like the Normal or Poisson distribution.
- ❖ The Chi-squared test allows you to assess your trained regression model's goodness of fit on the training, validation, and test data sets.
- ❖ **Karl Pearson introduced this test in 1900 for categorical data analysis and distribution. This test is also known as ‘Pearson’s Chi-Squared Test’.**
- ❖ Chi-Squared Tests are most commonly used in hypothesis testing. A hypothesis is an assumption that any given condition might be true, which can be tested afterwards.
- ❖ The Chi-Square test estimates the size of inconsistency between the expected results and the actual results when the size of the sample and the number of variables in the relationship is mentioned.

- ❖ These tests use degrees of freedom to determine if a particular null hypothesis can be rejected based on the total number of observations made in the experiments. Larger the sample size, more reliable is the result.
- ❖ There are two main types of Chi-Square tests namely –
 - ❖ Independence
 - ❖ Goodness-of-Fit

Independence

- ❖ The Chi-Square Test of Independence is a derivable statistical test which observes whether the two sets of variables are likely to be related with each other or not.
- ❖ This test is used when we have counts of values for two nominal or categorical variables and is considered as non-parametric test.
- ❖ A relatively large sample size and independence of observations are the required criteria for conducting this test.

For Example-

- ❖ In a movie theatre, suppose we made a list of movie genres.
- ❖ Let us consider this as the first variable. The second variable is whether or not the people who came to watch those genres of movies have bought snacks at the theatre.
- ❖ Here the null hypothesis is that the genre of the film and whether people bought snacks or not are unreliable.
- ❖ If this is true, the movie genres don't impact snack sales.

Goodness-Of-Fit

- ❖ In statistical hypothesis testing, the Chi-Square Goodness-of-Fit test determines whether a variable is likely to come from a given distribution or not.
- ❖ To test whether an observed distribution matches an expected distribution. We can use this test when we have value counts for categorical variables.
- ❖ This test demonstrates a way of deciding if the data values have a “good enough” fit for our idea or if it is a representative sample data of the entire population.

Scenario

A company wants to analyze the relationship between employees' education level and their job satisfaction. The data is presented in a contingency table that categorizes employees based on their education levels (High School, Bachelor's, and Master's) and their job satisfaction levels (Satisfied, Neutral, and Dissatisfied). Using the Chi-Square Test for Independence, determine whether there is a statistically significant relationship between education level and job satisfaction.

Education Level	Satisfied	Neutral	Dissatisfied
High School	50	30	20
Bachelor's	40	25	35
Master's	30	20	50

Scenario

Testing Relationship Between Education Level and Job Satisfaction
We have a contingency table showing the counts of employees' satisfaction levels (Satisfied, Neutral, Dissatisfied) across three education levels (High School, Bachelor's, Master's)

Education Level	Satisfied	Neutral	Dissatisfied
High School	50	30	20
Bachelor's	40	25	35
Master's	30	20	50

Import Required Libraries

1. `import numpy as np`
2. `from scipy.stats import chi2_contingency`

❖ `chi2_contingency`: This function from the `scipy.stats` library is used to perform the Chi-Square Test for Independence.

Define the Contingency Table

1. `data = [`
2. `[50, 30, 20], # High School`
3. `[40, 25, 35], # Bachelor's`
4. `[30, 20, 50] # Master's`
5. `]`

- ❖ The data represents the observed frequencies.
- ❖ Rows represent categories of one variable (Education Level: High School, Bachelor's, Master's).
- ❖ Columns represent categories of another variable (Job Satisfaction: Satisfied, Neutral, Dissatisfied).

Perform the Chi-Square Test

1. `chi2, p, dof, expected = chi2_contingency(data)`

- ❖ `chi2_contingency(data)`:
- ❖ Takes the contingency table as input.
- ❖ Returns:
- ❖ **Chi-Square statistic (chi2)**: Measures the difference between observed and expected frequencies.
- ❖ **p-value (p)**: Probability of observing the data assuming the null hypothesis is true.
- ❖ **Degrees of Freedom (dof)**: Calculated as $(rows-1) \times (columns-1)$
- ❖ **Expected frequencies (expected)**: Frequencies expected if the variables are independent

Display Results

1. `print("Chi-Square Test Results:")`
2. `print(f"Chi-Square Statistic: {chi2:.4f}")`
3. `print(f"Degrees of Freedom: {dof}")`
4. `print(f"P-Value: {p:.4f}")`
5. `print("\nExpected Frequencies:")`
6. `print(expected)`

❖ Outputs the Chi-Square statistic, degrees of freedom, p-value, and the expected frequencies.

Interpret the Results

1. `alpha = 0.05` # Significance level
2. `if p < alpha:`
3. `print("\nResult: Reject the null hypothesis (variables are dependent).")`
4. `else:`
5. `print("\nResult: Fail to reject the null hypothesis (variables are independent).")`

❖ Significance level (α): Usually set at 0.05.

❖ Hypotheses

❖ Null Hypothesis (H_0): The two variables are independent (no relationship).

❖ Alternative Hypothesis (H_1): The two variables are dependent (there is a relationship).

Scenario

Testing Relationship Between Education Level and Job Satisfaction
We have a contingency table showing the counts of employees' satisfaction levels (Satisfied, Neutral, Dissatisfied) across three education levels (High School, Bachelor's, Master's)

Education Level	Satisfied	Neutral	Dissatisfied
High School	50	30	20
Bachelor's	40	25	35
Master's	30	20	50

For mathematically calucution:

Calculate the Chi-Square Statistic

❖ The Chi-Square statistic (χ^2) is computed using the formula:

Where:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

- ❖ O: Observed frequency (from the input table).
- ❖ E: Expected frequency (calculated assuming the variables are independent).
- ❖ From the given contingency table

Education Level	Satisfied	Neutral	Dissatisfied	Row Totals
High School	50	30	20	100
Bachelor's	40	25	35	100
Master's	30	20	50	100
Column Totals	120	75	105	300

The expected frequencies are calculated as:

$$E_{ij} = \frac{\text{Row Total} \times \text{Column Total}}{\text{Grand Total}}$$

For example: **Expected value for High School & Satisfied:**

$$E = \frac{100 \times 120}{300} = 40$$

Repeat this for all cells.

Expected value for High School & Neutral, High School & Dissatisfied

And

Expected value for Bachelor's & Satisfied, Bachelor's & Neutral, Bachelor's & Dissatisfied

Expected value for Master's & Satisfied, Master's & Neutral, Master's & Dissatisfied

First Row (High School)

- $E_{11} = \frac{(100 \times 120)}{300} = 40.00$
- $E_{12} = \frac{(100 \times 75)}{300} = 25.00$
- $E_{13} = \frac{(100 \times 105)}{300} = 35.00$

Second Row (Bachelor's)

- $E_{21} = \frac{(100 \times 120)}{300} = 40.00$
- $E_{22} = \frac{(100 \times 75)}{300} = 25.00$
- $E_{23} = \frac{(100 \times 105)}{300} = 35.00$

Third Row (Master's)

- $E_{31} = \frac{(100 \times 120)}{300} = 40.00$
- $E_{32} = \frac{(100 \times 75)}{300} = 25.00$
- $E_{33} = \frac{(100 \times 105)}{300} = 35.00$

The expected table is:

Education Level	Satisfied	Neutral	Dissatisfied
High School	40.00	25.00	35.00
Bachelor's	40.00	25.00	35.00
Master's	40.00	25.00	35.00

Compute Chi-Square Statistic (χ^2):

The Chi-Square statistic is computed as:

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

For each cell:

First Row (High School)

- $\frac{(50-40)^2}{40} = \frac{(10)^2}{40} = \frac{100}{40} = 2.50$
- $\frac{(30-25)^2}{25} = \frac{(5)^2}{25} = \frac{25}{25} = 1.00$
- $\frac{(20-35)^2}{35} = \frac{(-15)^2}{35} = \frac{225}{35} = 6.43$

Second Row (Bachelor's)

- $\frac{(40-40)^2}{40} = \frac{(0)^2}{40} = 0.00$
- $\frac{(25-25)^2}{25} = \frac{(0)^2}{25} = 0.00$
- $\frac{(35-35)^2}{35} = \frac{(0)^2}{35} = 0.00$

Third Row (Master's)

- $\frac{(30-40)^2}{40} = \frac{(-10)^2}{40} = \frac{100}{40} = 2.50$
- $\frac{(20-25)^2}{25} = \frac{(-5)^2}{25} = \frac{25}{25} = 1.00$
- $\frac{(50-35)^2}{35} = \frac{(15)^2}{35} = \frac{225}{35} = 6.43$

Summing all these values:

$$\chi^2 = 2.50 + 1.00 + 6.43 + 0.00 + 0.00 + 0.00 + 2.50 + 1.00 + 6.43 = \mathbf{19.86}$$

Thus, the Chi-Square statistic is:

$$\chi^2 = \mathbf{19.86}$$

Compute Degrees of Freedom

The degrees of freedom (dof) is given by:

$$\text{dof} = (\text{Number of Rows} - 1) \times (\text{Number of Columns} - 1)$$

For this table:

$$\text{dof} = (3 - 1) \times (3 - 1) = 2 \times 2 = 4$$

Compute the P-Value

- ❖ The p-value is determined by finding the area under the Chi-Square distribution curve to the right of the calculated χ^2 statistic (16.5714) with 4 degrees of freedom.
- ❖ This is calculated using the Chi-Square cumulative distribution function (CDF):

$$\text{p-value} = 1 - \text{CDF}(\chi^2, \text{dof})$$

- ❖ Using a statistical table or software:

$$\text{p-value} = 1 - \text{CDF}(19.86, 4) = 0.0054$$

```
import scipy.stats as stats
# Parameters
x = 19.86 # Value
df = 4
# Degrees of freedom
#Compute the CDF
chi_square_cdf = stats.chi2.cdf(x, df)
chi_square_cdf
```

0.999467798668642

(df)	.99	.975	.95	.9	.1	.05	.025	.01
1	-----	0.001	0.004	0.016	2.706	3.841	5.024	6.635
2	0.020	0.051	0.103	0.211	4.605	5.991	7.378	9.210
3	0.115	0.216	0.352	0.584	6.251	7.815	9.348	11.345
4	0.297	0.484	0.711	1.064	7.779	9.488	11.143	13.277
5	0.554	0.831	1.145	1.610	9.236	11.070	12.833	15.086
6	0.872	1.237	1.635	2.204	10.645	12.592	14.449	16.812
7	1.239	1.690	2.167	2.833	12.017	14.067	16.013	18.475
8	1.646	2.180	2.733	3.490	13.362	15.507	17.535	20.090
9	2.088	2.700	3.325	4.168	14.684	16.919	19.023	21.666
10	2.558	3.247	3.940	4.865	15.987	18.307	20.483	23.209
11	3.053	3.816	4.575	5.578	17.275	19.675	21.920	24.725
12	3.571	4.404	5.226	6.304	18.549	21.026	23.337	26.217
13	4.107	5.009	5.892	7.042	19.812	22.362	24.736	27.688
14	4.660	5.629	6.571	7.790	21.064	23.685	26.119	29.141
15	5.229	6.262	7.261	8.547	22.307	24.996	27.488	30.578
16	5.812	6.908	7.962	9.312	23.542	26.296	28.845	32.000
17	6.408	7.564	8.672	10.085	24.769	27.587	30.191	33.409
18	7.015	8.231	9.390	10.865	25.989	28.869	31.526	34.805
19	7.633	8.907	10.117	11.651	27.204	30.144	32.852	36.191
20	8.260	9.591	10.851	12.443	28.412	31.410	34.170	37.566
21	8.897	10.283	11.591	13.240	29.615	32.671	35.479	38.932

Decision:

- ❖ Since the p-value (0.0054) is less than the significance level (0.05), we reject the null hypothesis, concluding that education level and job satisfaction are related.

Student's t-Test

- ❖ Student's t-test is a statistical test used to compare the means of two groups to determine if they are significantly different from each other. It is commonly used when the sample size is small (typically less than 30) and the population standard deviation is unknown..
- ❖ We can define the Student t-test as a method that tells you how significant the differences can be between different groups.
- ❖ **A Student t-test is defined as a statistic and this is used to compare the means of two different populations.**
- ❖ It is a method that is often used in hypothesis testing to find out whether a process or whether a given treatment actually has any effect on the population of interest, or whether or not two populations are different from each other.
- ❖ You can test the difference between these two groups with the help of the Student t-test.
- ❖ The null hypothesis (H_0) is one that tells the true difference between these groups.
- ❖ The alternate hypothesis (H_a) is one that tells the true difference is different from zero.

Student t Test Introduction

- ❖ In the year 1908, an Englishman named **William Sealy Gosset** developed the t-test as well as t distribution.
- ❖ **William** worked at the Guinness brewery in Dublin and found which existing statistical techniques using large samples were not useful for the small sample sizes which he encountered in his work.
- ❖ The **t distribution** belonging under a family of curves in which the number of degrees of freedom specifies a particular curve.

Purpose of Student's t-Test

- ❖ To compare the means of two groups and determine if they are significantly different.
- ❖ To test hypotheses about population means using sample data.
- ❖ To assess the impact of an intervention or treatment in experiments.

Student t-Test Formula

- ❖ The formula for the two-sample t-test / the Student's t-test) is shown below.

$$\text{Student t Test Formula, } t = \frac{\overline{x_1} - \overline{x_2}}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

- ❖ In the formula given above, **t** is equal to the t-value, **x1** and **x2** are the means of the two groups being compared, **s²** is the **pooled standard error** of the two groups, and **n1** and **n2** are the numbers of observations in each of the groups.
- ❖ A larger t-value denotes the difference between group means is greater than the pooled standard error, indicating a more significant difference between the groups.
- ❖ You can compare your calculated t-value against the values in a critical value chart to determine whether your t-value is greater than what would be expected by chance.
- ❖ If so, you can reject the null hypothesis and you can conclude which two groups are in fact different.

Types of Student t-Test

❖ When choosing a Student t-test, two things need to be kept in mind: whether the groups being compared are coming from a single population or two different populations, There are different types of t-tests, but the two most common ones are.

1. Independent Samples T-Test

2. Paired Samples T-Test

1. Independent Samples T-Test:

❖ This test is used when comparing the means of two independent groups. For example, comparing the average scores of two different groups of participants in an experiment or comparing the means of two different treatment groups.

❖ The formula for the t-statistic in the independent samples t-test is:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

❖ Where $\bar{X}_1$ and $\bar{X}_2$ are the sample means s_1 and s_2 are the sample standard deviations, and n_1 and n_2 are the sample sizes.

Paired Samples T-Test:

- ❖ This test is used when comparing the means of two related groups. For example, comparing the scores of the same group of participants before and after a treatment.
- ❖ The formula for the t-statistic in the paired samples t-test is:

$$t = \frac{\bar{d}}{\frac{s_d}{\sqrt{n}}}$$

- ❖ where $\bar{d}$ is the mean of the differences between paired observations, s_d is the standard deviation of the differences, and n is the number of pairs.

- ❖ A company wants to evaluate the effectiveness of its training program on employee performance. To assess this, the performance scores of 8 employees were recorded both before and after the training. Using the Student's t-test, determine whether the training program has led to a statistically significant improvement in employee performance.
- ❖ Given Performance Scores:
- ❖ Before Training: [65, 70, 68, 72, 75, 78, 74, 71]
- ❖ After Training: [78, 82, 80, 85, 87, 90, 86, 83]
- ❖ Perform a paired t-test and interpret the results.

❖ **A company seeks to evaluate the effectiveness of its training program on employee performance. To achieve this, performance scores of 8 employees are recorded both before and after the training. The objective is to determine whether the training program has led to a statistically significant improvement in employee performance.**

❖ This program evaluates whether a training program has significantly improved employee performance using a paired t-test. The paired t-test compares the means of two related groups (before and after training) to check for significant differences.

❖ Here Assume the Performance Scores

❖ `before_training = [65, 70, 68, 72, 75, 78, 74, 71]`

❖ `after_training = [78, 82, 80, 85, 87, 90, 86, 83]`

❖ Two lists represent performance scores before and after training for 8 employees.

❖ Each index in `before_training` corresponds to the same employee's score in `after_training`.

❖ **Import Required Libraries**

1. `import numpy as np`
2. `from scipy import stats`

❖ `numpy` is imported for numerical computations (though not explicitly used here). `scipy.stats` is imported to perform the paired t-test (`ttest_rel` function).

❖ **Define the Data (Performance Scores)**

1. `before_training = [65, 70, 68, 72, 75, 78, 74, 71]`
2. `after_training = [78, 82, 80, 85, 87, 90, 86, 83]`

❖ Two lists represent performance scores before and after training for 8 employees.

❖ Each index in `before_training` corresponds to the same employee's score in `after_training`.

❖ **Perform the Paired t-Test**

1. `t_stat, p_value = stats.ttest_rel(before_training, after_training)`

❖ `ttest_rel()` is used for paired data (same employees before and after training).

❖ This function returns:

❖ `t_stat`: The t-statistic, a measure of difference between means.

❖ `p_value`: The probability that the difference is due to random chance.

❖ **Print the t-Test Results**

1. `print(f"T-statistic: {t_stat:.3f}")`

2. `print(f"P-value: {p_value:.3f}")`

❖ Interpret the Results

1. $\alpha = 0.05$
2. if $p_value < \alpha$:
3. `print("Reject the null hypothesis: The training significantly improved performance.")`
4. else:
5. `print("Fail to reject the null hypothesis: No significant improvement due to training.")`

❖ $\alpha = 0.05$ (5% significance level) is a common threshold in hypothesis testing.

❖ If $p_value < 0.05$, we reject the null hypothesis, concluding significant improvement.

❖ Otherwise, we fail to reject the null hypothesis, meaning no strong evidence that training improved performance.

- ❖ **Mathematical Calculation for Paired t-Test**
- ❖ We will now manually compute the paired t-test step by step using the given data.
- ❖ **Given Data (Performance Scores)**

Employee	Before Training (Xi)	After Training (Yi)	Difference (Di = Yi–Xi)
1	65	78	13
2	70	82	12
3	68	80	12
4	72	85	13
5	75	87	12
6	78	90	12
7	74	86	12
8	71	83	12
Sum	573	671	96

❖ Calculate the Mean Difference

$$\bar{D} = \frac{\sum D_i}{n}$$

$$\bar{D} = \frac{96}{8} = 12$$

❖ Calculate the Standard Deviation of Differences

The formula for the standard deviation of the differences (s_D) is:

$$s_D = \sqrt{\frac{\sum (D_i - \bar{D})^2}{n - 1}}$$

❖ Calculate $(D_i - D^-)^2$ for Each Employee

Employee	D_i	$(D_i - D^-)$	$(D_i - D^-)^2$
1	13	$13 - 12 = 1$	1
2	12	$12 - 12 = 0$	0
3	12	$12 - 12 = 0$	0
4	13	$13 - 12 = 1$	1
5	12	$12 - 12 = 0$	0
6	12	$12 - 12 = 0$	0
7	12	$12 - 12 = 0$	0
8	12	$12 - 12 = 0$	0
Sum	96	-	2

$$s_D = \sqrt{\frac{2}{8-1}} = \sqrt{\frac{2}{7}} = \sqrt{0.2857} \approx 0.535$$

❖ Calculate the t-Statistic

The paired t-test formula is:

$$t = \frac{\bar{D}}{s_D / \sqrt{n}}$$

Substituting the values:

$$t = \frac{12}{\frac{0.535}{\sqrt{8}}}$$

$$t = \frac{12}{0.1892} \approx 63.44$$

- ❖ **Determine the p-value and Conclusion**
- ❖ From the t-table, for degrees of freedom ($df = n - 1 = 8 - 1 = 7$) at a significance level of 0.05, the critical t-value is 2.447. Since $t = 63.44$ is much greater than 2.447, the p-value is very small ($p < 0.001$), leading to rejection of the null hypothesis.

A **t-table** provides critical values for different degrees of freedom and confidence levels. Below is an excerpt:

df	t (0.10, two-tailed)	t (0.05, two-tailed)	t (0.01, two-tailed)
6	1.943	2.447	3.707
7	1.895	2.447	3.499
8	1.860	2.306	3.355

Since our $df = 7$ and $\alpha = 0.05$, we find $t = 2.447$.

- ❖ $df = n - 1 = 8 - 1 = 7$ Since we have 8 employees, the degrees of freedom (df) = 7.

- ❖ Compare t-Statistic with t-Critical
- ❖ t-Statistic from our calculation: 63.44
- ❖ t-Critical (from table): 2.447
- ❖ Since $63.44 > 2.447$, we reject the null hypothesis and conclude a significant difference.

❖ Instead of using a t-table manually, we can compute the t-critical value using Python:

1. `from scipy.stats import t`
2. `# Define degrees of freedom`
3. `df = 7`
4. `# Compute t-critical for a two-tailed test with  $\alpha = 0.05$`
5. `t_critical = t.ppf(1 - 0.05 / 2, df)`
6. `print(f"T-critical value: {t_critical:.3f}")`

❖ **Output:**

❖ **T-critical value: 2.447**

Central limit theorem and its application, confidence intervals

- ❖ Central Limit Theorem (CLT) The Central Limit Theorem (CLT) states that, regardless of the shape of the population distribution, the sampling distribution of the sample mean will be approximately normal if the sample size is sufficiently large (typically $n \geq 30$).
- ❖ **Key points about the CLT:**
- ❖ **Large sample size:** The CLT is most accurate when the sample size is large, typically considered to be 30 or more samples.
- ❖ **Independent samples:** The samples taken must be independent of each other.
- ❖ **Population mean and variance:** The CLT allows us to estimate the population mean and variance using the sample mean and variance, even if the original population distribution is not normal.

- ❖ **How would you use hypothesis testing to determine if a new drug is more effective than an existing one?**
- ❖ **Hypothesis Testing for Comparing Drug Effectiveness**
- ❖ To determine if a new drug is more effective than an existing one, we use hypothesis testing. The process follows these steps:
- ❖ **Define Hypotheses**
- ❖ Null Hypothesis (H_0): The new drug is not more effective than the existing drug (i.e., both have the same effect or the new drug is worse).
- ❖ Alternative Hypothesis (H_1): The new drug is more effective than the existing drug
- ❖ **Select a Significance Level (α)**
- ❖ Common choices: 0.05 (5%) or 0.01 (1%)

❖ **Collect Data**

- ❖ Conduct a randomized controlled trial (RCT) with two groups:
 - ❖ Treatment Group: Receives the new drug.
 - ❖ Control Group: Receives the existing drug (or placebo, if applicable).
- ❖ Measure an outcome variable (e.g., symptom reduction, recovery rate).

❖ **Choose the Appropriate Statistical Test**

- ❖ If comparing mean effectiveness (e.g., symptom reduction scores): Use a t-test (independent or paired).
- ❖ If comparing proportions (e.g., recovery rates): Use a Chi-square test or Z-test for proportions.

❖ **Compute the Test Statistic and p-Value**

- ❖ If $p\text{-value} < \alpha$, reject $H_0 \rightarrow$ The new drug is significantly more effective.
- ❖ If $p\text{-value} \geq \alpha$, fail to reject $H_0 \rightarrow$ No significant evidence that the new drug is better.

❖ **Interpret the Results**

- ❖ If significant, the new drug may be considered for approval.
- ❖ If not significant, more studies may be needed.

A company claims that their battery lasts an average of 50 hours. A researcher tests a sample of 30 batteries and finds an average life of 48 hours with a standard deviation of 5 hours. Formulate the null and alternative hypotheses for this study.

❖ **Solution:**

❖ **Hypothesis Testing for Battery Life Claim**

❖ A company claims that its batteries last an average of 50 hours. A researcher tests a sample of 30 batteries and finds that their average lifespan is 48 hours, with a standard deviation of 5 hours. We will perform a hypothesis test to determine if the sample provides enough evidence to reject the company's claim.

❖ **Step 1: Define Hypotheses**

❖ We use a two-tailed t-test since we want to check if the actual battery life is different from the claimed 50 hours.

❖ Null Hypothesis (H_0): The average battery life is 50 hours

❖ $H_0 : \mu = 50$

- ❖ Alternative Hypothesis (H_1): The average battery life is not 50 hours.

$$H_1 : \mu \neq 50$$

❖ Step 2: Given Data

- ❖ Sample size (n) = 30
- ❖ Sample mean ($\bar{x}$) = 48 hours
- ❖ Sample standard deviation (s) = 5 hours
- ❖ Claimed mean (μ_0) = 50 hours
- ❖ Significance level (α) = 0.05

❖ Step 3: Calculate the Test Statistic (t-score)

- ❖ The t-score formula is:

$$t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}}$$

❖ Substituting the values:

$$t = \frac{48 - 50}{\frac{5}{\sqrt{30}}}$$

$$t = \frac{-2}{0.9129} = -2.19$$

❖ **Step 4: Compute the p-value**

❖ Since this is a two-tailed test, we calculate:

$$\text{❖ } p = 2 \times P(T > |t|)$$

❖ Using a t-distribution with degrees of freedom ($df = n - 1 = 29$),

❖ we get:

$$\text{❖ } p \approx 0.0366$$

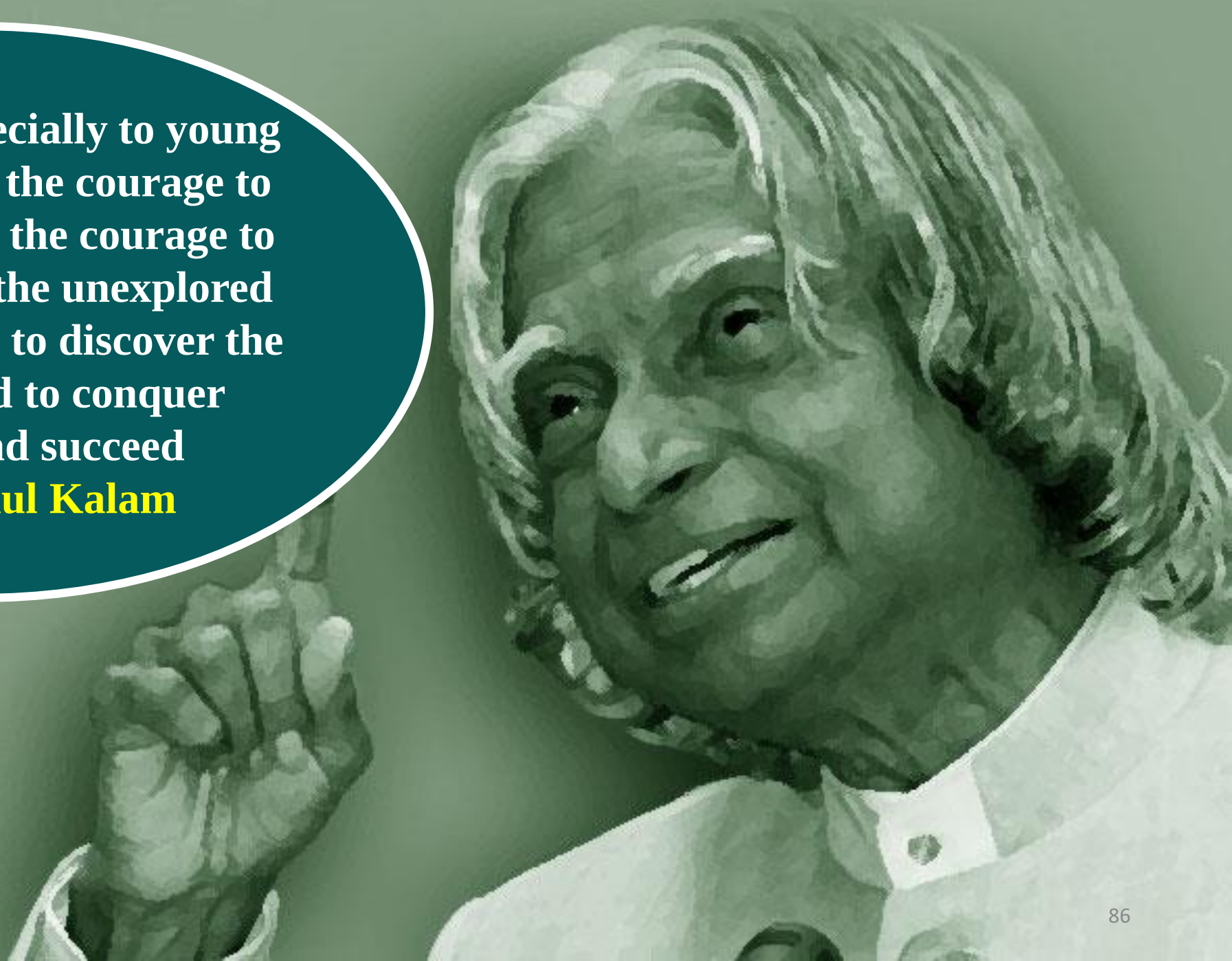
❖ Step 5: Decision Rule

- ❖ If $p\text{-value} < \alpha$ (0.05), reject H_0 .
- ❖ If $p\text{-value} \geq \alpha$ (0.05), fail to reject H_0 .
- ❖ Since $p\text{-value}$ (0.0366) < 0.05 , we reject the null hypothesis.

❖ Step 6: Conclusion

- ❖ There is statistically significant evidence that the average battery life is different from the claimed 50 hours. The sample suggests that the actual battery life is lower than what the company claims.

My message, especially to young people is to have the courage to think differently, the courage to invent, to travel the unexplored path, the courage to discover the impossible and to conquer problems and succeed
A. P. J. Abdul Kalam



A red pushpin is pinned to the top center of a yellow sticky note.

Thank
You!